

Observation limits and compilation checks

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The main result is a checkable distinction between what is measured and what is assumed. The studies below are complete within their stated scope. They do not establish quantum advantage or a new general theorem.

Exact finite mathematics

In a fixed 4,096-state repetition X-error model, the same complete syndrome statistics can support opposite decoder rankings. A two-atom perturbation stays inside the earlier single-rate intervals. The exact minimum total-variation distance to a tie is **0.0022031647031184**. At the radius there is a tie; a slightly larger change reverses the ranking.

Adding a known-preparation logical outcome identifies the observed risk comparison. Hidden fault correlations can remain underdetermined. One selected pair becomes identifiable if exact single means are separately known; the other ten selected pairs still admit explicit counterexamples.

Recorded hardware data and calibration

A fresh replay reproduced **1.8 million parity bits from 200,000 measured Google shots**. These previously exposed data are not a fresh holdout. Across 36 settings and 72,000 empirical-law resamples, no decoder improvement was resolved. Selective labeling used exactly the same physical shots as full paired-data reuse. Label-cost savings are hypothetical; physical-shot savings are zero.

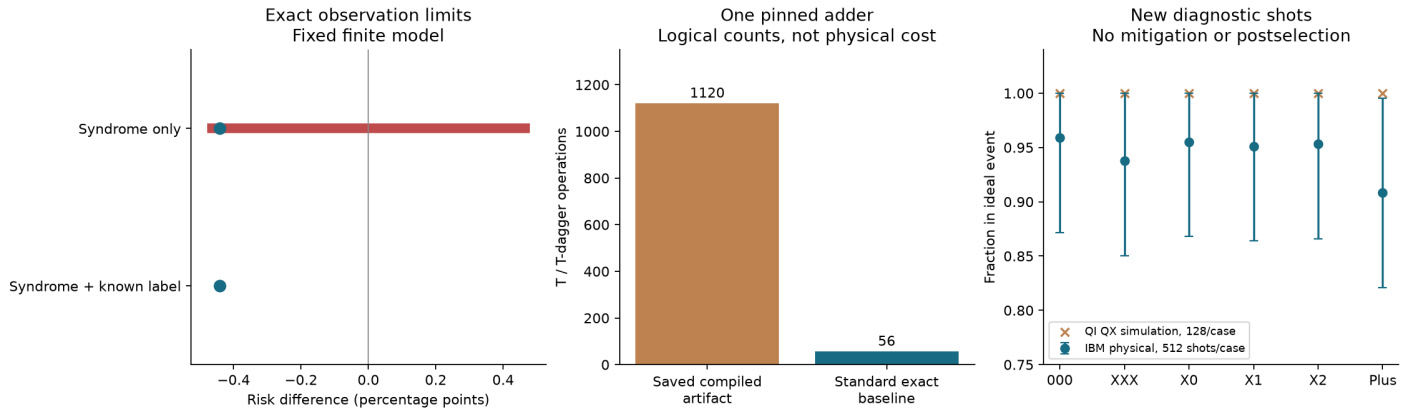
New provider experiments

IBM ibm_fez: six circuits, 3,072 physical shots, 3 charged QPU seconds. **Quantum Inspire:** six QX jobs, 768 simulated shots; physical devices were unavailable. All shots were retained. These are wiring and preparation/readout checks, not validation of the two-round fault law.

Compilation audit and review gate

A pinned public PNNL adder artifact contains **1,120 T operations**. A standard exact seven-T/T-dagger Toffoli baseline uses **56** for the same original circuit. The exact primitive identities were checked algebraically. This is known decomposition machinery, not a new compiler.

At one trace-aligning global phase, the saved artifact has numerical operator residual 0.2031076762. Its configured example still returns the correct full-register basis state with probability 0.9998993047. Good basis probabilities alone do not establish coherent equivalence. The artifact-specific synthesis tolerance remains undocumented.



How this was checked

A separate exact Python checker rebuilt 28 envelopes and 2,688 fiber records, with corruption rejection. Independent MATLAB observation checks passed on Z890, Z790, GE63, Mac mini M4 and Red Hat VM. RTX 5090 and 4090 independently replayed all 1,024 compiled columns. The separate MATLAB resource result and any harness failures are retained in the package.

What should happen next

Obtain expert criticism of the observation assumptions and ask the public artifact authors which generating settings apply. Do not interpret the 20:1 logical gate-count ratio as physical resource savings. Novelty, journal suitability and human acceptance remain open.

Sources: [Google data](#); [FTCircuitBench](#); [DOE roadmap](#). Full assumptions, source hashes, proofs and replay instructions are in PAPER.md and VALIDATION.md. Roadmap alignment does not imply DOE affiliation or funding.